In this page, I investigate some typical finite groups.
Specifically, investigate;
Abelian and Abelization
Commutator and Derived subgroup
Upper-Central Series and Solvablity and Composition Series
Center and Conjugacy classes.
Explicit all subgroup

The general Properties of S_n, D_n, C_n will be covered in other page.

$\mathbb{Z}_2 \times S_3$, Conjugacy Classes.

Since $|G:G_a| = |C_a|$, find the Centralizer of $(0, (12))$.

Since $(1,2)$ commutes only id and itself, and $\mathbb{Z}_2$ is abelian,

$$C_G(0,(1,2)) = \{(0,(12)), (0, id), (1,(1,2)), (1, id)\}.$$

Therefore, the Conjugacy classes of $(0,(12))$ has $2 \cdot 6/4 = 3$ elements.

Since $\mathbb{Z}_2$ is abelian (or 0 is identity), the first term never change, and since the Conjugate cannot change the cycle structure, the conjugacy classes of $(0, (12))$ is

$$\{(0,(12)), (0,(13)), (0,(23))\},$$

and similarly, the conjugacy classes of $(1,(12))$ is

$$\{(1,(12)), (1,(13)), (1,(23))\},$$

" $(0,(123))$ is

$$\{(0,(123)), (0,(132))\}$$

" $(1,(123))$ is

$$\{(1,(123)), (1,(132))\}$$

and $(0, id)$ and $(1, id)$ have conjugacy classes only containing itself.

$\sqrt{S_3 \times S_3}$, Conjugacy classes.

Let $(1\ 2) \times (1\ 2) \in S_3 \times S_3$, $(1\ 2)$ commute only id or itself, thus

$$C_G((1\ 2)) = \{(id, id), (id, (1\ 2)), ((1\ 2), id), ((1\ 2), (1\ 2))\},$$

So, the conjugacy classes of $(1\ 2) \times (1\ 2)$ has 9 elements

S_4.

$S_4 =$	$\{ id,$
1,1,2	$(1\ 2), (1\ 3), (1\ 4), (2\ 3), (2\ 4), (3\ 4),$
2,2	$(1\ 2)(3\ 4), (1\ 3)(2\ 4), (1\ 4)(2\ 3),$
1,3	$(1\ 2\ 3), (1\ 3\ 2), (1\ 3\ 4), (1\ 4\ 3), (1\ 2\ 4), (1\ 4\ 2), (2\ 3\ 4), (2\ 4\ 3),$
4	$(1\ 2\ 3\ 4), (1\ 2\ 4\ 3), (1\ 3\ 2\ 4), (1\ 3\ 4\ 2), (1\ 4\ 2\ 3), (1\ 4\ 3\ 2)\}$
Partition	

A_4. Conjugacy classes.

$A_4 = \{id, (1\,2)(3\,4), (1\,3)(2\,4), (1\,4)(2\,3),$
$(1\,2\,3), (1\,3\,2), (1\,2\,4), (1\,4\,2), (1\,3\,4), (1\,4\,3),$
$(2\,3\,4), (2\,4\,3)\}$

Since the conjugate cannot change the cycle type, the conjugacy class of each element in A_4 are contained the set of same cycle type. But we don't know it exactly equals to the set, let's calculate.

$$(1\,2\,3)(1\,2)(3\,4)(1\,2\,3)^{-1} = (2\,3)(1\,4)$$

$$(1\,2\,4)(1\,2)(3\,4)(1\,2\,4)^{-1} = (2\,4)(3\,1)$$

therefore, the conjugacy classes of $(1\,2)(3\,4)$ equals the set of $(2,2)$ cycles.

$$(1\,3\,2)(1\,2\,3)(1\,3\,2)^{-1} = (3\,1\,2) = (1\,2\,3) \quad (\text{In actually, } (1\,2\,3)^{-1} = (1\,3\,2))$$

$$(1\,2\,4)(1\,2\,3)(1\,2\,4)^{-1} = (2\,4\,3)$$

$$(1\,4\,2)(1\,2\,3)(1\,4\,2)^{-1} = (4\,1\,3) = (1\,3\,4)$$
$$(1\,3\,4)(1\,2\,3)(1\,3\,4)^{-1} = (3\,2\,4) = (2\,4\,3)$$
$$(1\,4\,3)(1\,2\,3)(1\,4\,3)^{-1} = (4\,2\,1) = (1\,4\,2)$$
$$(2\,3\,4)(1\,2\,3)(2\,3\,4)^{-1} = (1\,3\,4)$$
$$(2\,4\,3)(1\,2\,3)(2\,4\,3)^{-1} = (1\,4\,2)$$
$$(1\,2)(3\,4)(1\,2\,3)((1\,2)(3\,4))^{-1} = (2\,1\,4)$$
$$(1\,3)(2\,4)(1\,2\,3)(1\,3)(2\,4) = (3\,4\,1) = (1\,3\,4)$$
$$(1\,4)(2\,3)(1\,2\,3)(1\,4)(2\,3) = (4\,3\,2) = (2\,4\,3)$$

therefore, the conjugacy classes of $(1\,2\,3)$ has just four elements.

Generally, since $|G : G_a| = C_a$ and $G_a = C_G(a)$ in the conjugate action, therefore number of elements of the orbit is $|G|/|C_G(a)|$.

Since $C_{A_4}((1\,2\,3)) = \{id, (1\,2\,3), (1\,3\,2)\}$,

the orbit of $(1\,2\,3)$ has $12/3 = 4$ elements.

ΓA_5. The smallest group of non-abelian simple group.

「Klein 4-group.

Let $G = \mathbb{Z}_2 \times \mathbb{Z}_2$ be the Klein 4-group.

Then, a map $\varphi : G \to S_4$;

$(0,0) \mapsto id$, $(1,0) \mapsto (1\,2)(3\,4)$, $(0,1) \mapsto (1\,3)(2\,4)$, $(1,1) \mapsto (1\,4)(2\,3)$

is Embedding